INCH-POUND

MIL-S-13949/2C
6 June 1994
SUPERSEDING
MIL-P-13949/2B
11 February 1987

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL GB (WOVEN E-GLASS REINFORCEMENT, MAJORITY POLYFUNCTIONAL EPOXY RESIN, HOT STRENGTH RETENTION, METAL CLAD OR UNCLAD)

Inactive for new design after 6 June 1994
For new design use MIL-S-13949/5.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: G in accordance with MIL-S-13949.

Resin system: B in accordance with MIL-S-13949.

Special consideration (third letter designator): N or P in accordance with MIL-S-13949.

Type of metal foil: Any as specified in accordance with MIL-S-13949.

Weight of metal foil: 1, 2, 3, 4, 5 or 0 in accordance with MIL-S-13949.

Solderability: Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: C or X in accordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Flammability: Not applicable.

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm)).

Crosswise: 50,000 pounds per square inch minimum.

Lengthwise: 60,000 pounds per square inch minimum.

Lengthwise at elevated temperature: 30,000 pounds per square inch minimum.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Peel strength values for copper foil: See table I.

TABLE I. Peel strength values, all base material thicknesses.

Test and copper foil weight	Pounds per inch of width
After thermal stress:	
1-oz/ft^2 (designator 1)	8.0
2-oz/ft^2 (designator 2)	11.0
3-oz/ft^2 (designator 3)	12.0
4-oz/ft^2 (designator 4)	12.0
5-oz/ft^2 (designator 5)	12.0
At elevated temperature of 150°C:	
1-oz/ft^2 (designator 1)	3.0
2-oz/ft^2 (designator 2)	4.0
3-oz/ft^2 (designator 3)	5.0
4-oz/ft^2 (designator 4)	5.0
5-oz/ft^2 (designator 5)	5.0
After exposure to processing solutions:	
1-oz/ft^2 (designator 1)	7.0
2-oz/ft^2 (designator 2)	9.0
3-oz/ft^2 (designator 3)	12.0
4-oz/ft^2 (designator 4)	12.0
5-oz/ft^2 (designator 5)	12.0

Arc resistance: 60 seconds minimum.

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 40 kilovolts minimum.

Electric strength: Not applicable.

Loss tangent at 1 MHz: 0.030 maximum.

Permittivity at 1 MHz: 5.4 maximum.

Surface resistivity:

After moisture resistance conditioning: 10^4 megohm minimum.

After elevated temperature (150°C) conditioning: 10^3 megohm minimum.

Volume resistivity:

After moisture resistance conditioning: 10^6 megohm-cm minimum.

After elevated temperature (150°C) conditioning: 10^5 megohm-cm minimum.

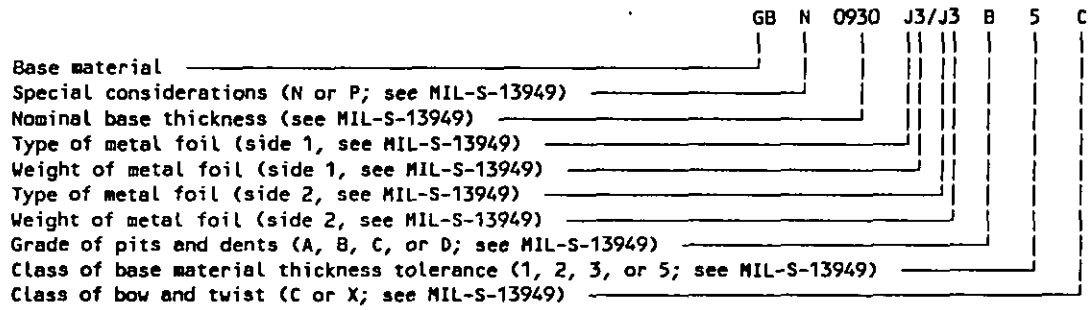
Water absorption:

Not applicable for base material thickness less than .020 inch (0.51 mm).
Base material thickness of .020 inch (0.51 mm) to .031 inch (0.79 mm): 0.80 percent maximum.
Base material thickness of .032 inch (0.81 mm) to .062 inch (1.58 mm): 0.35 percent maximum.
Base material thickness of .063 inch (1.60 mm) to .093 inch (2.36 mm): 0.25 percent maximum.
Base material thickness of .094 inch (2.38 mm) to .125 inch (3.18 mm): 0.20 percent maximum.
Base material thickness of .126 inch (3.20 mm) to .250 inch (6.35 mm): 0.13 percent maximum.

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Supersession information: Not applicable.

Type designation: Consists of "GB" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, ME, MI
Navy - AS, MC, OS, SH
Air Force - 11, 19, 99
DLA - ES
NS

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0050)

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